

E-COMMERCE SALES PERFORMANCE ANALYSIS USING SQL

Prepared By

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Project: E-Commerce Sales Performance Analysis Using SQL

Tools Used: MySQL, Microsoft Excel, GitHub

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Project Objective

The objective of this project is to analyse e-commerce sales data using SQL and identify key business insights related to profitability, shipping efficiency, delivery performance, and sales trends. The analysis helps understand business performance and supports data-driven decision-making.

Business Questions

Q1. Which Product Categories Are Performing Below and Above the Overall Business Profit Margin?

Objective:

Compare the profit margin of each product category with the overall business profit margin to identify high-performing and low-performing categories.

Business Value:

Helps management focus on profitable categories and improve underperforming product segments.

Q2. How Does Shipping Cost Affect Profit Margins Across Different Shipping Modes?

Objective:

Analyze the relationship between shipping costs and profit margins for each shipping mode.

Business Value:

Identifies cost-efficient shipping methods and helps optimize logistics expenses.

Q3. Which Sub-Categories Are Causing Margin Leakage Within Low-Performing Categories?

Objective:

Identify sub-categories that generate low profit margins and contribute to profit leakage within weak-performing categories.

Business Value:

Enables businesses to review pricing, cost structures, and product strategies for loss-making sub-categories.

Q4. Does Longer Delivery Time Negatively Impact Profitability or Priority Orders?

Objective:

Examine whether longer shipment durations are associated with lower profitability and order priorities.

Business Value:

Provides insights into operational efficiency and helps improve delivery performance and customer satisfaction.

Q5. What Are the Yearly and Monthly Trends in Sales, Profit, and Profit Margin?

Objective:

Analyze sales, profit, and profit margin trends over different years and months.

Business Value:

Helps identify seasonality, business growth patterns, and supports forecasting and strategic planning.

Tools Used

- MySQL
 - Microsoft Excel
 - GitHub
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SQL Concepts Applied

- Aggregate Functions (SUM(), AVG(), COUNT())
- GROUP BY
- ORDER BY
- CASE WHEN
- Subqueries

- Common Table Expressions (CTEs)
 - Date Functions (YEAR(), MONTH(), STR_TO_DATE())
 - Profit Margin Calculations
 - Business Performance Analysis
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Expected Outcomes

- Identify profitable and low-performing categories.
- Understand the impact of shipping costs on profitability.
- Detect margin leakage at the sub-category level.
- Evaluate delivery efficiency and its effect on business performance.
- Discover yearly and monthly sales and profit trends for strategic decision-making.